Solutions of Equations in One Variable

Newton's Method

Numerical Analysis (9th Edition)

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Beamer Presentation Slides
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Outline

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1. Newton's Method: Derivation
2. Example using Newton's Method & Fixed-Point Iteration

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Newton's Method

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Newton's (or the *Newton-Raphson*) method is one of the most powerful and well-known numerical methods for solving a root-finding problem.

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Various ways of introducing Newton's method

- Graphically, as is often done in calculus.
- As a technique to obtain faster convergence than offered by other types of functional iteration.
- Using Taylor polynomials. We will see there that this particular derivation produces not only the method, but also a bound for the error of the approximation.

Newton's Method

Derivation

Newton's Method

Derivation

- Suppose that $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to p such that $f'(p_0) \neq 0$ and $|p - p_0|$ is "small."

Newton's Method

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- Consider the first Taylor polynomial for $f(x)$ expanded about p_0 and evaluated at $x = p$.

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)),$$

where $\xi(p)$ lies between p and p_0.

Newton's Method

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where $\xi(p)$ lies between p and p_0.

- Since $f(p) = 0$, this equation gives

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)).$$

Newton's Method

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Derivation (Cont'd)

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- Newton's method is derived by assuming that since $|p - p_0|$ is small, the term involving $(p - p_0)^2$ is much smaller, so

$$0 \approx f(p_0) + (p - p_0)f'(p_0).$$

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- Solving for p gives

$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} \equiv p_1.$$

Newton's Method

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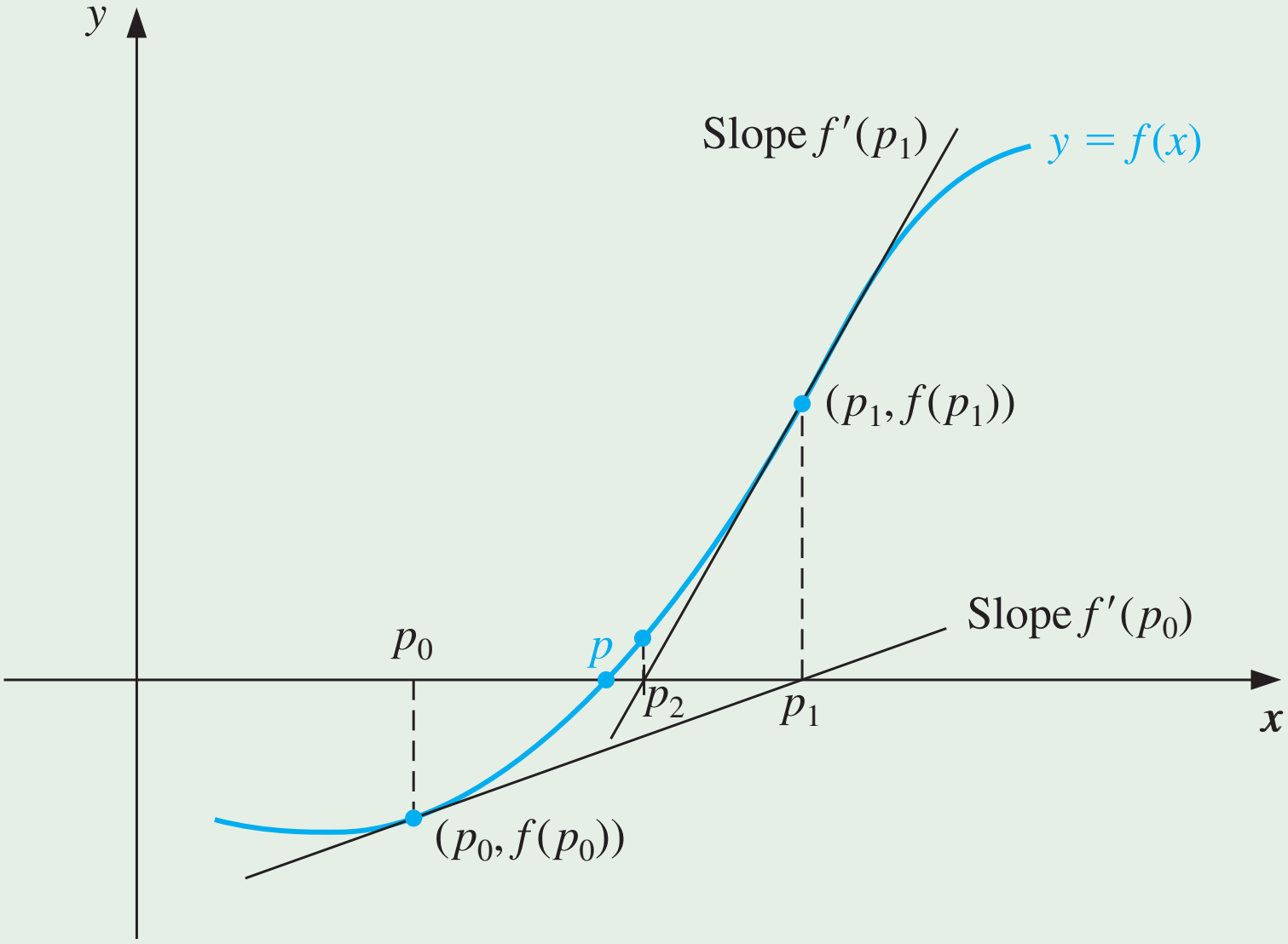
Newton's Method

This sets the stage for Newton's method, which starts with an initial approximation p_0 and generates the sequence $\{p_n\}_{n=0}^{\infty}$, by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$

Newton's Method: Using Successive Tangents



Newton's Algorithm

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6. Output p

Newton's Method

Stopping Criteria for the Algorithm

- Various stopping procedures can be applied in Step 3.3.

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$$|p_N - p_{N-1}| < \epsilon \tag{1}$$

$$\frac{|p_N - p_{N-1}|}{|p_N|} < \epsilon, \quad p_N \neq 0, \quad \text{or} \tag{2}$$

$$|f(p_N)| < \epsilon \tag{3}$$

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- Note that none of these inequalities give precise information about the actual error $|p_N - p|$.

Newton's Method as a Functional Iteration Technique

Functional Iteration

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- Newton's Method

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Newton's Method

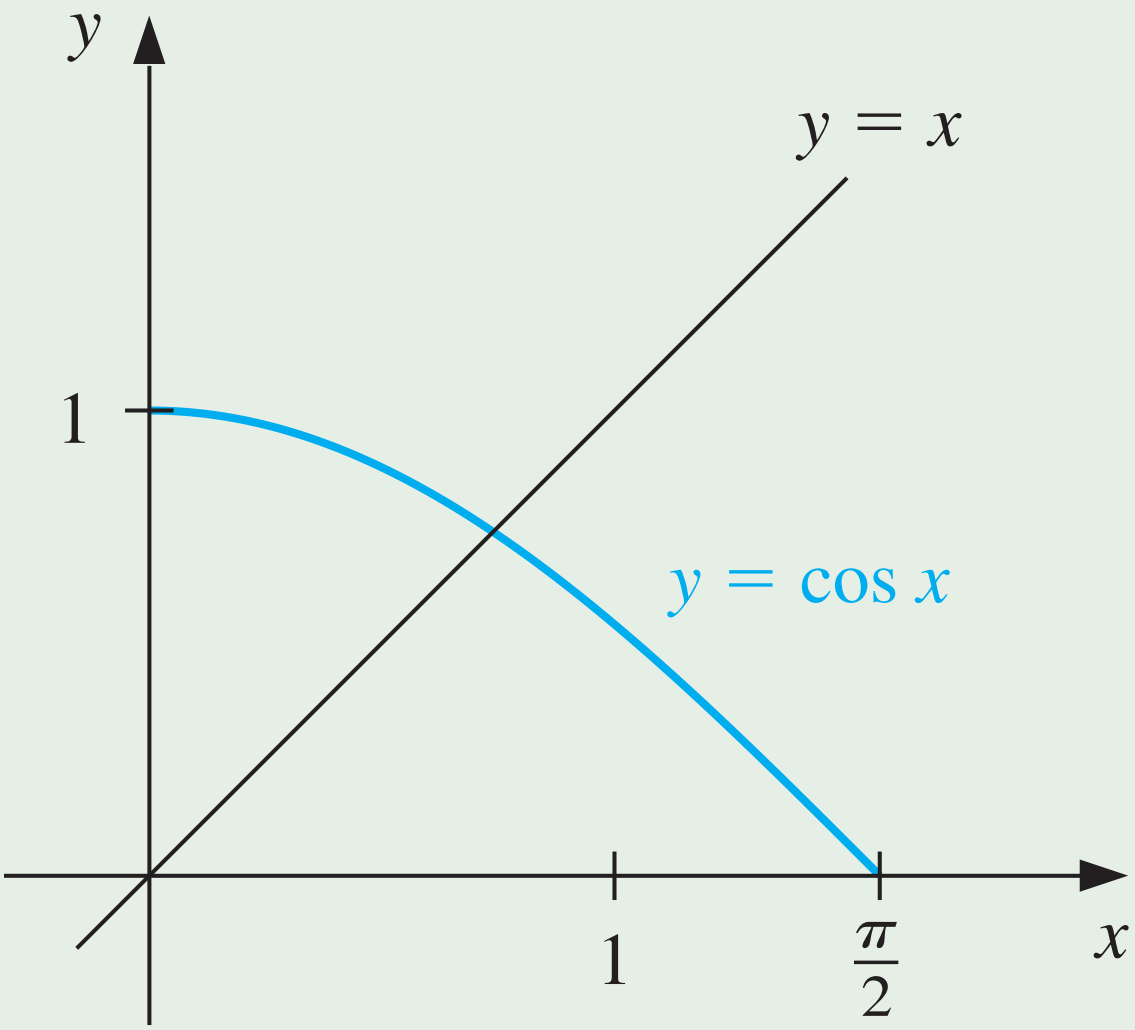
Example: Fixed-Point Iteration & Newton's Method

Consider the function

$$f(x) = \cos x - x = 0$$

Approximate a root of f using (a) a fixed-point method, and (b) Newton's Method

Newton's Method & Fixed-Point Iteration



Newton's Method & Fixed-Point Iteration

(a) Fixed-Point Iteration for $f(x) = \cos x - x$

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- A solution to this root-finding problem is also a solution to the fixed-point problem

$$x = \cos x$$

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- The following table shows the results of fixed-point iteration with $p_0 = \pi/4$.

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  and the graph implies that a single fixed-point p lies in $[0, \pi/2]$.
- The following table shows the results of fixed-point iteration with $p_0 = \pi/4$.
- The best conclusion from these results is that $p \approx 0.74$.

Newton's Method & Fixed-Point Iteration

Fixed-Point Iteration: $x = \cos(x)$, $x_0 = \frac{\pi}{4}$

n	p_{n-1}	p_n	$\lvert p_n - p_{n-1}\rvert$	e_n/e_{n-1}
1	0.7853982	0.7071068	0.0782914	—
2	0.707107	0.760245	0.053138	0.678719
3	0.760245	0.724667	0.035577	0.669525
4	0.724667	0.748720	0.024052	0.676064
5	0.748720	0.732561	0.016159	0.671826
6	0.732561	0.743464	0.010903	0.674753
7	0.743464	0.736128	0.007336	0.672816

Newton's Method

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- This gives the approximations shown in the following table.

Newton's Method

Newton's Method for $f(x) = \cos(x) - x$, $x_0 = \frac{\pi}{4}$

n	p_{n-1}	$f(p_{n-1})$	$f'(p_{n-1})$	p_n	$\lvert p_n - p_{n-1} \rvert$
1	0.78539816	-0.078291	-1.707107	0.73953613	0.04586203
2	0.73953613	-0.000755	-1.673945	0.73908518	0.00045096
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- Because of the agreement of p_3 and p_4 we could reasonably expect this result to be accurate to the places listed.

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Convergence using Newton's Method

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- This will clearly be false unless p_0 is a good approximation to p.
- If p_0 is not sufficiently close to the actual root, there is little reason to suspect that Newton's method will converge to the root.
- However, in some instances, even poor initial approximations will produce convergence.

Convergence using Newton's Method

Convergence Theorem for Newton's Method

Convergence using Newton's Method

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- Let $f \in C^2[a, b]$. If $p \in (a, b)$ is such that $f(p) = 0$ and $f'(p) \neq 0$.
- Then there exists a $\delta > 0$ such that Newton's method generates a sequence $\{p_n\}_{n=1}^{\infty}$, defined by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f(p'_{n-1})}$$

converging to p for any initial approximation

$$p_0 \in [p - \delta, p + \delta]$$

Convergence using Newton's Method

Convergence Theorem (1/4)

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Convergence Theorem (1/4)

- The proof is based on analyzing Newton's method as the functional iteration scheme $p_n = g(p_{n-1})$, for $n \geq 1$, with

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- Let k be in $(0, 1)$. We first find an interval $[p - \delta, p + \delta]$ that g maps into itself and for which $|g'(x)| \leq k$, for all $x \in (p - \delta, p + \delta)$.
- Since f' is continuous and $f'(p) \neq 0$, part (a) of Exercise 29 in Section 1.1 Ex 29 implies that there exists a $\delta_1 > 0$, such that $f'(x) \neq 0$ for $x \in [p - \delta_1, p + \delta_1] \subseteq [a, b]$.

Convergence using Newton's Method

Convergence Theorem (2/4)

Convergence using Newton's Method

Convergence Theorem (2/4)

- Thus g is defined and continuous on $[p-\delta_1, p+\delta_1]$. Also

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

for $x \in [p-\delta_1, p+\delta_1]$, and, since $f \in C^2[a,b]$, we have $g \in C^1[p-\delta_1, p+\delta_1]$.

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  for $x \in [p-\delta_1, p+\delta_1]$, and, since $f \in C^2[a,b]$, we have $g \in C^1[p-\delta_1, p+\delta_1]$.
- By assumption, $f(p) = 0$, so

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

Convergence using Newton's Method

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Convergence Theorem (3/4)

- Since g' is continuous and $0 < k < 1$, part (b) of Exercise 29 in Section 1.1 Ex 29 implies that there exists a δ, with $0 < \delta < \delta_1$, and

$$|g'(x)| \leq k, \quad \text{for all } x \in [p - \delta, p + \delta].$$

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- It remains to show that g maps $[p-\delta, p+\delta]$ into $[p-\delta, p+\delta]$.
- If $x \in [p-\delta, p+\delta]$, the Mean Value Theorem MVT implies that for some number ξ between x and p, $|g(x) - g(p)| = |g'(\xi)||x-p|$. So
$$|g(x) - p| = |g(x) - g(p)| = |g'(\xi)||x-p| \le k|x-p| < |x-p|.$$

Convergence using Newton's Method

Convergence Theorem (4/4)

Convergence using Newton's Method

Convergence Theorem (4/4)

- Since $x \in [p - \delta, p + \delta]$, it follows that $|x - p| < \delta$ and that $|g(x) - p| < \delta$. Hence, g maps $[p - \delta, p + \delta]$ into $[p - \delta, p + \delta]$.

Convergence using Newton's Method

Convergence Theorem (4/4)

- Since $x \in [p-\delta, p+\delta]$, it follows that $|x-p| < \delta$ and that $|g(x)-p| < \delta$. Hence, g maps $[p-\delta, p+\delta]$ into $[p-\delta, p+\delta]$.
- All the hypotheses of the Fixed-Point Theorem Theorem 2.4 are now satisfied, so the sequence $\{p_n\}_{n=1}^{\infty}$, defined by

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad \text{for } n \geq 1,$$

converges to p for any $p_0 \in [p-\delta, p+\delta]$.

Outline

1. Newton's Method: Derivation
2. Example using Newton's Method & Fixed-Point Iteration
3. Convergence using Newton's Method
4. Final Remarks on Practical Application

Newton's Method in Practice

Choice of Initial Approximation

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Newton's Method in Practice

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Newton's Method in Practice

Choice of Initial Approximation

- The convergence theorem states that, under reasonable assumptions, Newton's method converges provided a sufficiently accurate initial approximation is chosen.
- It also implies that the constant k that bounds the derivative of g, and, consequently, indicates the speed of convergence of the method, decreases to 0 as the procedure continues.
- This result is important for the theory of Newton's method, but it is seldom applied in practice because it does not tell us how to determine δ.

Newton's Method in Practice

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Newton's Method in Practice

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- an initial approximation is selected
- and successive approximations are generated by Newton's method.
- These will generally either converge quickly to the root,
- or it will be clear that convergence is unlikely.

Questions?

Reference Material

Exercise 29, Section 1.1

Let $f \in C[a, b]$, and let p be in the open interval (a, b).

Exercise 29 (a)

Suppose $f(p) \neq 0$. Show that a $\delta > 0$ exists with $f(x) \neq 0$, for all x in $[p - \delta, p + \delta]$, with $[p - \delta, p + \delta]$ a subset of $[a, b]$.

Return to Newton's Convergence Theorem (1 of 4)

Exercise 29 (b)

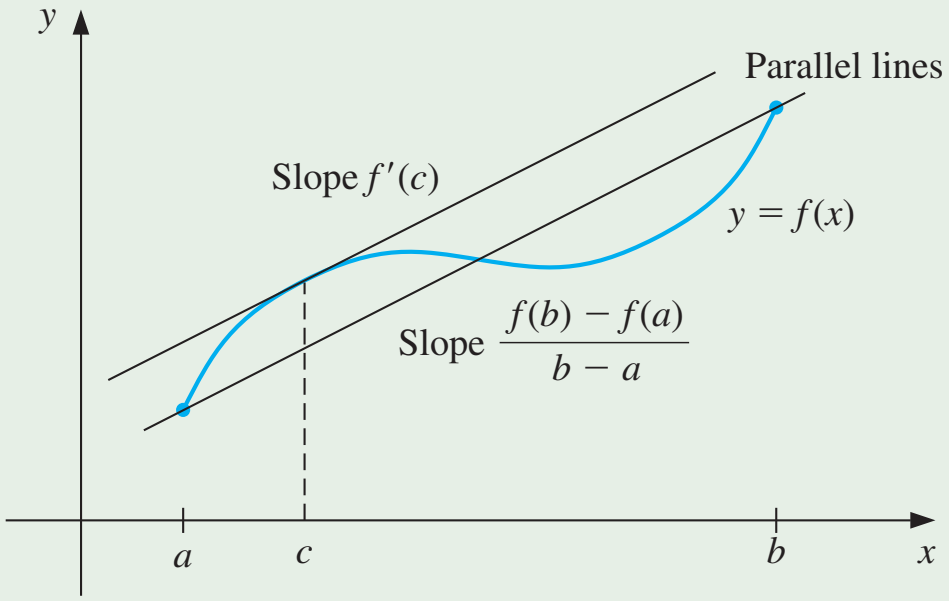
Suppose $f(p) = 0$ and $k > 0$ is given. Show that a $\delta > 0$ exists with $|f(x)| \leq k$, for all x in $[p - \delta, p + \delta]$, with $[p - \delta, p + \delta]$ a subset of $[a, b]$.

Return to Newton's Convergence Theorem (3 of 4)

Mean Value Theorem

If $f \in C[a, b]$ and f is differentiable on (a, b), then a number c exists such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Return to Newton's Convergence Theorem (3 of 4)

Fixed-Point Theorem

Let $g \in C[a, b]$ be such that $g(x) \in [a, b]$, for all x in $[a, b]$. Suppose, in addition, that g' exists on (a, b) and that a constant $0 < k < 1$ exists with

$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b).$$

Then for any number p_0 in $[a, b]$, the sequence defined by

$$p_n = g(p_{n-1}), \quad n \geq 1,$$

converges to the unique fixed point p in $[a, b]$. $\square$

Return to Newton's Convergence Theorem (4 of 4)